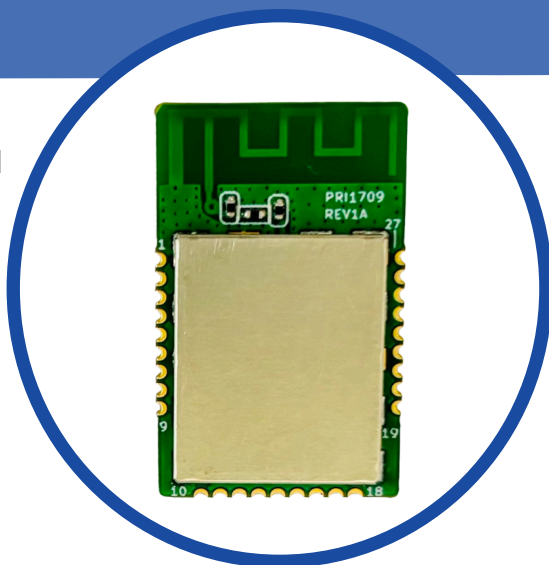


# Zigbee MG24 Module NLN811P

**NEBULAE**<sup>®</sup>  
DESIGN TO CONNECT

*The Intelligent, Secure, and Energy-Efficient IoT Connectivity Solution for Next-Gen IoT devices Manufacturer.*

NLN811P is based on the Silicon Labs EFR32MG24 Wireless SoCs, ideal for mesh IoT wireless connectivity using Matter, Open Thread and Zigbee. With key features like high performance 2.4 GHz RF, low current consumption, an AI/ML hardware accelerator and Secure Vault, IoT device makers can create smart, robust, and energy-efficient products that are secure from remote and local cyber-attacks.



## Key Features

- **Processor:** Silicon Labs EFR32MG24 SoC
  - **CPU:** 78 MHz ARM Cortex-M33
  - **Memory:** Up to 1536 kB Flash, 256 kB SRAM
  - **Frequency Bands:** 2.4 GHz ISM
  - **Performance:** TX Power up to +19.5 dBm
  - **RF Sensitivity:** -105.4 dBm @ 250 kbps
  - **Security:** Crypto, AES-128/192/256 (HW Accelerated)
- **Firmware:** OTA updates, Secure Boot
- **Network Protocols:** Matter, Bluetooth 5, Zigbee, Thread, Multi Protocol
- **Power & Current:** Supply: 3.3V, Standby: 50  $\mu$ A, TX: 160 mA, RX: 6 mA
- **Physical:**
  - **Size:** 26.7 × 16 × 3.6 mm
  - **I/O:** Serial, Analog, Digital (21 pins)
  - **Mount:** 1.27mm Pitch SMD
  - **Shielding:** EMI RF shield
- **Environment:**
  - **Temp Range:** -20°C to +85°C

## Benefits

- Longer battery life for IoT devices
- Future-Proof Protocols
- Flexibility for diverse IoT applications.
- High Wireless Range & Reliability
- Suitable for Complex Tasks and Local Processing
- Robust protection against various cyber threats

## Applications

- Energy management systems
- Smart meters and utility monitoring
- Remote controls & voice assistants
- Building Automation
- Door locks & access control systems
- Smart switches, plugs, and outlets
- Smart hubs & gateways



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